Han Wang

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Profile

CTO and Co-founder of Yumi Health, building an AI-native electronic health record for skilled nursing facilities. Research at Berkeley Artificial Intelligence Research (BAIR) and Berkeley DeepDrive (BDD) spans generative modeling, human–AI interaction, and multi-agent reinforcement learning for autonomous driving.

Professional Experience

Yumi Health | CTO and Co-founder

- Co-founded Yumi Health to build an AI-native electronic health record for skilled nursing. Helped recruit the founding engineering team after a $25M angel round and launched the pilot after three weeks of on-site development with nursing home clinicians.
- Built Yumi's first clinical copilot for iOS, integrating live voice and patient charts. Developed an agent that prepares orders, notes, and assessments, asks for missing information, and requires clinician approval before saving to the chart.
- Built configurable workflows for admission assessments, order sets, and care planning.
- Built an in-house retrieval pipeline for Minimum Data Set (MDS) assessments that finds question-specific chart evidence and links recommendations to their sources for clinician review. Added Medicare reimbursement previews.

Layr | Chief AI Scientist
Previously AI Research Scientist at Platflow (now Layr)

- Led development and production deployment of a fine-tuned 12B Gemma model for skilled nursing assessments and diagnosis coding.
- Developed Layr's in-house MDS retrieval and evaluation tools, ranking chart evidence within question-specific lookback windows and benchmarking suggested answers against reference assessments.
- Shipped daily hospitalization risk reports for 52 skilled nursing facilities, with automated source validation and an access-controlled dashboard.
- Built a PointClickCare workflow that uses diagnoses and chart evidence to flag missing care plan interventions and lets nurses review and apply suggestions.

Education

University of California, Berkeley — *GPA 3.93/4.0* — **CA, United States**
Ph.D. in Civil and Environmental Engineering — *Advisor: Maria Laura Delle Monache* — Expected Dec 2026
Concentration: Transportation Engineering
M.S. in Electrical Engineering and Computer Sciences — *Advisor: Alexandre Bayen* — Expected Dec 2026
M.Eng. in Transportation Engineering — *Advisor: Alexander Skabardonis* — 08/2020 – 05/2021

Southeast University — *GPA 85.75/100* — **Nanjing, China**
M.Eng. in Transportation Engineering — *Advisor: Jian Zhang, Bin Ran* — 09/2017 – 05/2020

Southwest Jiaotong University — *GPA 80.27/100* — **Chengdu, China**
B.S. in Transportation Engineering — 09/2013 – 06/2017

Selected Berkeley coursework: Deep Reinforcement Learning, Responsible GenAI and Decentralized Intelligence, Computer Vision, Advanced Control, and Immersive Computing and Virtual Reality.

Skills

Programming Languages:
Python, Swift, C#

Backends:
MySQL, AWS, Azure

Frameworks and Libraries:
PyTorch, Ray, Gym, MuJoCo, OpenCV, Ollama, Unsloth, LangChain

Simulation and Modeling:
Unity, SteamVR, Blender, SUMO

Research Experience

Perception, Prediction and Control of Collision Indeterminacy via Stochastic Generation **06/2023-06/2025**
*Predicting the motion of AV with the **pre-trained image generative model**.* *Funded by Berkeley Deep Drive and Allstate*
• Role: Data fusion and comprehensive perception based on the onboard sensor array. Occupancy image dataset collecting, generative decoder pre-training, end-to-end training with perception encoder, the building of the simulation platform, and the design & training of the AV controller in the simulator.
• Tools: ML Model Design & Training: PyTorch, Ray | Simulation: SUMO | Dataset: Argoverse
• Outcome: Developed a probabilistic occupancy risk assessment (PORA) metric, validated through a scenario-generation-based microsimulation platform, demonstrated to be effective in predicting 87% of the conflicts that could not be predicted by TTC-2.

Congestion Impacts Reduction via CAV-in-the-loop Lagrangian Energy Smoothing (CIRCLES) **03/2022-12/2024**
*World's largest open-track fieldtest, sending **100 auto-vehicle(AV)s** to I-24 during the morning peak to improve traffic.*
Funded by Toyota, Nissan, and General Motors. Supported by Tennessee Department of Transportation
• Role: Led the centralized speed planner to publish the optimized speed for all 100 AVs in real-time during the experiment.
• Tools: Reinforcement Learning Speed Planner: PyTorch, Ray, Gym | Backend: MySQL, Python | API: PHP
• Outcome: From the field test data, we observed an 8% improvement overall, with 7% and 10% improvements upstream and downstream, respectively, and a 52% improvement during congestion formation at the bottleneck.

Understanding Resilience and Equity of Transportation System During Hazardous Events **07/2021-12/2024**
*An **equity-focused resilience framework** for urban transportation.* *Funded by California Department of Transportation*
• Role: Designed a social-physical integrated evaluation index to quantify resilience and equity, integrating social equity considerations with system resilience to ensure equitable recovery for underserved communities. Formulated recovery strategies as a Markov Decision Process (MDP) and implemented reinforcement learning (RL) algorithms using Actor-Critic methods. Conducted large-scale simulations of the San Francisco Bay Area transportation network, leveraging GMNS and OpenStreetMap data to validate the framework's scalability and effectiveness.
• Tools: Network Modeling: GMNS, OpenStreetMap | Simulation: Python, TensorFlow | RL Algorithm: Actor-Critic
• Outcome: Achieved a significant reduction in inequity penalties, enhancing recovery performance by 13.8% over capacity-based strategies and 26.3% over population-based strategies. Results demonstrated effective prioritization and equitable recovery under various hazard scenarios.

Creating an Inclusive Bicycle Level of Service: Virtual Bicycle Simulator study **01/2021-08/2023**
*Gather user feedback on biking environments with **VR simulator** survey.* *Funded by California Department of Transportation*
• Role: Built the VR biking simulator, including hardware & software integration. Rebuilt the California streets in 3D scenarios.
• Tools: Simulator Development: Unity, C#, SteamVR | Modeling: Blender
• Outcome: The feedback from the survey proved the VR simulator provided a more realistic and engaging experience for participants, which enhanced their ability to make informed decisions about their route preferences.

Opportunities and Challenges for Runtime Behavior Generation in Games and Simulations **09/2023-06/2024**
***Runtime behavior generation** using large language models in game/simulation development.* *Personal Project*
• Role: Developed the runtime code compile module. Implemented the semantic search and prompt engineering of the agent. Develop the element tree system of test scenarios. Conducted generalization tests in various scenarios.
• Tools: System Development: Unity, C# | LLM: OpenAI API
• Outcome: The pressure test demonstrated the system in various game scenarios, achieving an 85% success rate. Thematic analysis of developer feedback indicates the improved dynamic game experience by integrating real-time behavior generation.

V2X-based Driving Perception Assistance System **09/2023-12/2024**
*Integration of **foundation models and AR** to enhance driving assistance systems via real-time visual aids* *Personal Project*
• Role: Designed and implemented a system integrating multimodal data (Lidar, cameras, maps, traffic updates) for real-time driving assistance. Developed AR visual aids for 3D object tracking and scene reconstruction. Utilized specialized LLMs for script generation and natural language interaction.
• Tools: System Development: Unity, C#, OpenCV | LLM: Ollama, LiteLLM
• Outcome: Achieved accurate 3D object tracking and reconstruction, enhancing driver situational awareness and safety through real-time AR visualizations. Refined functionality through human-in-loop interactions and driver feedback. Integrated foundation models for natural language command interpretation and script generation, improving adaptability and usability.

Research Experience (continued)

Vision-based Browser Automation using GenAI **01/2024-12/2024**
*Automating browser workflows using **multimodal LLM agents**.* *Personal Project*
• Role: Finetuned open-source LLMs with RAG for browser operation subtasks. Developed vision + HTML webpage scraper. Defined and implemented multi-agent workflow. Develop the browser operation API kit for LLM. Developed the Chrome extension for human-in-loop expert data collecting. Created the open dataset on HuggingFace.
• Tools: Finetune: Unsloth | Agent: Ollama, LangChain, LiteLLM
• Outcome: Enhanced automation stability and adaptability to website layout changes, enabling complex workflow automation across multiple sites. Integrated real-time debugging and visualization of automation steps, facilitating efficient troubleshooting.

Crowdsourcing Perceived Hazardous Pedestrian Locations **09/2020-05/2021**
SafeTport: An IOS App allowing users to report and view traffic hazards and crashes. *Funded by SafeTREC*
• Role: Developed the front-end and back-end of the system independently.
• Tools: IOS APP: Swift, Google Map Platform | Backend: MySQL, AWS, Restful API
• Outcome: Created a functional application, identified marketing strategy in user engagement and data collection.

Research on Multi-junction Energy-saving Access Control of Intelligent Electric Vehicles **07/2017–12/2020**
Developed control models and simulations for energy-saving access control in intelligent electric vehicles. *Funded by SEU*
• Role: Established a multi-agent control model using improved MADDPG algorithm for multi-intersections urban arterial in a connected environment. Designed traffic and communication network simulations using SUMO, OMNET++.
• Tools: Controller: TensorFlow | Simulation: SUMO, OMNET++

Resilience Research for Extremely Surging Traffic of Huning (Shanghai-Nanjing) Highway **03/2017–12/2019**
Analyze and improve the resilience of the Huning Highway system under extremely large traffic flow
Funded by Joint Research Institute on Internet of Mobility: Southeast University and University of Wisconsin
• Role: Led the establishment of the SUMO micro-simulation scenario for the entire Huning Highway. Designed the ramp control algorithm based on deep reinforcement learning.
• Tools: Simulation: SUMO | Ramp Controller: keras

Course Design of the Traffic Analysis I Driven by Big Data **01/2019–12/2019**
Enhanced traditional traffic analysis course content with data mining skills. *Funded by Didi Chuxing Technology Co.*
• Role: Added data mining related skills to the traditional traffic analysis course content. Designed course paper requirements integrating data of the GAIA Initiative from Didi.

Thesis

Thesis Title: *Human-AI Interaction in Autonomous Vehicle Operation* **Expected Dec 2026**
Degree: Ph.D. in Civil and Environmental Engineering **University of California, Berkeley, CA**
Objective: To develop a cooperative mixed-autonomy framework where autonomous vehicles act as traffic-control agents, integrating hierarchical control, human behavior-aware probabilistic risk assessment, and vendor-agnostic safety certification. Validated through theory and field experiments, demonstrating improved traffic throughput, reduced bottleneck density, and fewer conflicts.

Thesis Title: *Reinforcement Learning for Adaptive Traffic Control in Mixed-Autonomy Environments* **Expected Dec 2026**
Degree: M.S. in Electrical Engineering and Computer Sciences (Plan II) **University of California, Berkeley, CA**
Objective: To develop a PDE-ODE mixed-autonomy traffic model and reinforcement-learning-based hierarchical speed planner, validated in simulation and the 100-vehicle I-24 field deployment.

Thesis Title: *Multi-Agent Reinforcement Learning Method for Arterial Traffic Signal Control* **May 2020**
Degree: M.Eng. in Transportation Engineering **Southeast University, Nanjing, China**
Objective: To address the complexity of coordinating multi-intersection arterial traffic signal control systems by implementing a reinforcement learning approach that effectively manages the delayed reward dynamics inherent in such environments.

Publications

Journal Articles

- **Wang, H.**, Yeo, Y., Paiva, A. R., Goodman, J. P., Utke, J., & Monache, M. L. D. (2025). Dynamic risk assessment for autonomous vehicles from spatio-temporal probabilistic occupancy heatmaps. Accident Analysis & Prevention, 222, 108226.
- Tang, F., **Wang, H.**, & Delle Monache, M. L. (2025). Strategizing equitable transit evacuations: A data-driven reinforcement learning approach. Transportation Research Part C: Emerging Technologies, 180, 105342.
- Lee, J. W., **Wang, H.(co-first)**, Jang, K., Hayat, A., Bunting, M., Alanqary, A., ... (64 authors) & Bayen, A. M. (2025). Traffic control via connected and automated vehicles: An open-road field experiment with 100 cavs. IEEE Control Systems, 45(1), 28-60.
- **Wang, H.**, Fu, Z., Lee, J., Matin, H. N. Z., Alanqary, A., Urieli, D., ... (17 authors) & Monache, M. L. D. (2025). Hierarchical speed planner for automated vehicles: A framework for lagrangian variable speed limit in mixed autonomy traffic. IEEE Control Systems, 45(1), 111-138.
- Ameli, M., McQuade, S., Lee, J. W., Bunting, M., Nice, M., **Wang, H.**, ... & Bayen, A. M. (2025). Designing, simulating, and performing the 100-av field test for the circles consortium: Methodology and implementation of the largest mobile traffic control experiment to date. IEEE Control Systems, 45(1), 139-155.
- Zhang, J., Dong, S., Li, Z., Ran, B., Li, R., & **Wang, H.** (2019). An eco-driving signal control model for divisible electric platoons in cooperative vehicle-infrastructure systems. IEEE Access, 7, 83277-83285.

Conference Papers

- (Best Paper Award) Jennings, N., **Wang, H.**, Li, I., Smith, J., & Hartmann, B. (2024, October). What's the Game, then? Opportunities and Challenges for Runtime Behavior Generation. In Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology (pp. 1-13).
- Veksler, Y., Hornstein, S., **Wang, H.**, Monache, M. L. D., & Urieli, D. (2024). Cooperative Cruising: Reinforcement Learning based Time-Headway Control for Increased Traffic Efficiency. Accepted by AAAI 2025. arXiv:2412.02520.
- **Wang, H.**, Nick Zinat Matin, H.,& Delle Monache, M. L. (2024, June). Reinforcement learning-based adaptive speed controllers in mixed autonomy condition. In 2024 European Control Conference (ECC) (pp. 01-06). IEEE.
- Fu, Z., Kreidieh, A. R., **Wang, H.**, Lee, J. W., Delle Monache, M. L., & Bayen, A. M. (2023, June). Cooperative driving for speed harmonization in mixed-traffic environments. In 2023 IEEE Intelligent Vehicles Symposium (IV) (pp. 1-8). IEEE.
- **Wang, H.**, Wu, H., Lu, J., Tang, F., & Delle Monache, M. L. (2023, September). Communication Optimization for Multi-agent Reinforcement Learning-based Traffic Control System with Explainable Protocol. In 2023 IEEE 26th International Conference on Intelligent Transportation Systems (ITSC) (pp. 6068-6073). IEEE.
- **Wang, H.**, & Delle Monache, M. L. (2022, July). Urban network resilience analysis and equity emphasized recovery based on reinforcement learning. In 2022 European Control Conference (ECC) (pp. 01-06). IEEE.
- Tang, F., Cheng, L., **Wang, H.**, Mao, P., & Jiang, J. (2019). Research on the Impact of Car-Hailing on Travel Mode Choice: Evidence from Chengdu, China. In CICTP 2019 (pp. 6134-6145).
- Dong, S. Y., Zhang, J., **Wang, H.**, Ran, B., & Tan, H. C. (2018, July). A speed guidance-based signal control method for divisible platoon in CVIS. In 2018 IEEE 8th Annual International Conference on CYBER Technology in Automation, Control, and Intelligent Systems (CYBER) (pp. 1403-1408). IEEE.
- Dong, S. Y., Zhang, J., Chen, T. Y., **Wang, H.**, & Ran, B. (2018, July). CVIS-Based Intersection Signal Control Model for Indivisible Platoons. In 18th COTA International Conference of Transportation Professionals (pp. 359-368). Reston, VA: American Society of Civil Engineers.

Pre-prints

- Sprinkle, J. (corresponding), **Wang, H.(co-first)**, et al. (2025). Energy-Saving Control of Freeway Traffic: Field Experiments and Results. Submitted to Nature.
- **Wang, H.**, Yeo, Y., Paiva, A. R., Utke, J., & Monache, M. L. D. (2025). Modular framework for uncertainty prediction in autonomous vehicle motion prediction within complex traffic scenarios. Submitted to Transportation Research Part C: Emerging Technologies. arXiv preprint arXiv:2501.16480.
- Chekroun, R., **Wang, H.**, Lee, J., Toromanoff, M., Hornauer, S., Moutarde, F., & Monache, M. L. D. (2024). Mesoscale Traffic Forecasting for Real-Time Bottleneck and Shockwave Prediction. Submitted to Transportation Research Part C: Emerging Technologies. arXiv preprint arXiv:2402.05663.